

Species-Aware Module Review: ADP-heptose Biosynthesis in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Commissioned topic:** `adp_heptose_biosynthesis` **Resolved local bucket:** KEGG `ppu00541` — "Biosynthesis of various nucleotide sugars" (11 primary + 26 candidate genes) **Correct module identity:** KEGG module **M00064**, "ADP-L-glycero-D-manno-heptose biosynthesis (sedoheptulose-7P $\Rightarrow$ ADP-LDmanHep)" **Date:** 2026-08-08

1. Executive summary

- The commissioned topic `adp_heptose_biosynthesis` is a narrow, well-defined 4-enzyme pathway (KEGG **M00064**) that makes **ADP-L-glycero- β -D-manno-heptose**, the activated sugar donor for the heptose residues of the LPS **inner core**. It is **not** the same as the broad KEGG bucket `ppu00541`/`map00541` ("Biosynthesis of various nucleotide sugars") from which the 26-gene candidate list was drawn. That bucket conflates ≥ 6 unrelated nucleotide-sugar pathways.
- **Only 3 of the 26 candidate genes belong to this module.** The other 23 are KDO, dTDP-rhamnose, GDP-mannose/fucose, UDP-GlcNAc and UDP-glucuronate genes that should be excluded from ADP-heptose satisfiability.
- **Steps 1–4 are covered with high confidence:** `gmhA` (PP_1323), bifunctional `hldE` (PP_4934, covering two steps), and `gmhB` (PP_0059). KEGG orthology and UniProt curated EC numbers agree.
- **The final step — the ADP-L-glycero-D-manno-heptose-6-epimerase HldD/GmhD (K03274, EC 5.1.3.20) — is a genome-confirmed GAP in KT2440.** Four independent authorities agree there is no *hldD*: KEGG (`link/ppu/K03274` $\rightarrow$ none), UniProt/UniRef (no KT2440 protein in the UniRef50 or UniRef90 cluster of the same-genus HldD *P. aeruginosa* PA3337/Q9HYQ8), NCBI RefSeq (zero heptose-6-epimerase / *hldD/gmhD/rfaD* records for txid160488), and — decisively — **tblastn of PA3337 against the KT2440 chromosome**

(AE015451.2 + 9 other assemblies) yields only a 29%-identity best hit (a *galE*), with no ortholog-level ORF anywhere in the 6-frame-translated genome. Method calibration confirms sensitivity: true PA↔PP orthologs score 80–92% (gmhA 91.8%, gmhB 80.0%). So the gene is genuinely absent, not merely un-annotated.

- **CORRECTION (Iteration 2):** PP_3206 is **not** the epimerase — KEGG K26183 defines it as *fdeJ* (naringenin/flavonoid degradation, map00946). Its PANTHER-subfamily match reflected the broad NAD-epimerase/dehydratase superfamily, not HldD specificity.
- **Interpretation:** steps 1–4 still make ADP-D-glycero-D-manno-heptose and the downstream WaaC/WaaF/WaaG/WaaP inner-core machinery is intact, so heptose is incorporated. Because the D→L epimerase is absent, either **(a)** KT2440 uses **D-glycero-D-manno-heptose (D,D-Hep) directly** in its core (no epimerization needed → step 5 `not_expected_in_target_taxon`, and the generic module endpoint is wrong for this organism), or **(b)** a **non-orthologous/novel epimerase** supplies the activity (→ new gene/GO request). Note *P. aeruginosa* uses L,D-Hep and *does* encode HldD, so this is a KT2440-specific feature. Curation call: **step 5 = gap / module_needs_revision**; the decisive open question is the **stereochemistry of KT2440 inner-core heptose** (needs LPS structural data).

2. Target-organism pathway definition

Process included (M00064): cytoplasmic conversion of the pentose-phosphate-pathway intermediate **D-sedoheptulose-7-phosphate** into the nucleotide sugar **ADP-L-glycero-β-D-manno-heptose**, in four enzymatic activities:

#	Reaction	Enzyme	EC	KO
1	Sedoheptulose-7-P → D-glycero-D-manno-heptose-7-P	GmhA (phosphoheptose isomerase)	5.3.1.28	K03271
2	Heptose-7-P → D-glycero-β-D-manno-heptose-1,7-bisP	HldE kinase domain	2.7.1.167	K03272 (or K21344)
3	Heptose-1,7-bisP → D-glycero-β-D-manno-heptose-1-P	GmhB (bisphosphate 7-phosphatase)	3.1.3.82	K03273
4	Heptose-1-P → ADP-D-glycero-β-D-manno-heptose	HldE adenylyltransferase domain	2.7.7.70	K03272 (or K21345)
5	ADP-D-glycero-heptose → ADP-L-glycero-D-manno-heptose	HldD/GmhD (6-epimerase)	5.1.3.20	K03274

Canonical pathway biochemically established in *E. coli* (Kneidinger et al. 2002,

[P 11751812](#)

McArthur et al. 2005,

[P 16030223](#)

Neighboring pathways to keep separate: - KDO biosynthesis (KdsD/KdsA/KdsC/KdsB) — the *other* LPS inner-core sugar (3-deoxy-D-manno-octulosonate). Distinct module (M00063), even though KEGG places it on the same [map00541](#) overview. - **LPS core assembly / transfer** — WaaC, WaaF heptosyltransferases and WaaP/WapP/WapQ heptose kinases *use* ADP-heptose but are downstream (map00540 LPS biosynthesis), not part of M00064. - **dTDP-L-rhamnose** (rmlA-D/rfbA-D), **GDP-D-rhamnose/GDP-fucose** (gmd/rmd), **UDP-GlcNAc** (glmU/rffE), **UDP-glucuronate** (galU/udg) — all separate nucleotide-sugar modules.

Alternate names: module M00064 "ADP-LDmanHep biosynthesis"; genes historically *rfaE* (=hldE), *yaeD* (=gmhB), *waaD/rfaD/htrM* (=hldD), *gmhD* (=hldD).

3. Expected step model (satisfiability call per step)

Step	Enzyme (KO)	KT2440 gene	Call	Basis
1	GmhA (K03271)	PP_1323	covered	KEGG ortholog + UniProt EC 5.3.1.28
2	HldE kinase (K03272)	PP_4934	covered	KEGG ortholog + UniProt EC 2.7.1.167 (bifunctional)
3	GmhB (K03273)	PP_0059	covered	KEGG ortholog + UniProt EC 3.1.3.82
4	HldE adenylyltransferase (K03272)	PP_4934	covered	same bifunctional protein, UniProt EC 2.7.7.70
5	HldD/GmhD epimerase (K03274)	<i>none — absent from genome</i>	gap / module_needs_revision (possibly <code>not_expected_in_target_taxon</code>)	no gene in KEGG/UniProt/RefSeq; genome tblastn best hit 29% (a <i>galE</i>) across all 10 KT2440 assemblies

Module verdict: 4 of 5 catalytic steps covered by 3 genes (through ADP-D-glycero-D-manno-heptose). The terminal D→L epimerase (step 5) is **genome-confirmed absent**, so ADP-L-glycero-D-manno-heptose is *not* demonstrably produced by an M00064-canonical route. The module is therefore **not satisfiable as defined** in KT2440. Whether this is a true biological difference (D,D-Hep core → epimerase not needed) or a novel epimerase is the key open question — resolvable by LPS structural analysis.

4. Candidate genes and evidence

High-confidence, in-module (promote to "covered")

- **PP_1323** **gmhA** (Q88N89, 195 aa) — phosphoheptose isomerase, EC 5.3.1.28, KEGG K03271. Also carries synonym *diaA* (DnaA-interacting); the GmhA/DiaA fold is shared, but the EC and KO assignment are unambiguous. **Direct KEGG+UniProt evidence.**
- **PP_4934** **hldE** / **rfaE** (Q88D93, 473 aa) — bifunctional D-β-D-heptose-7-P kinase (EC 2.7.1.167) + heptose-1-P adenylyltransferase (EC 2.7.7.70), KEGG K03272. Covers module steps 2 and 4. **Direct evidence.**
- **PP_0059** **gmhB** (Q88RS0, 175 aa) — D,D-heptose-1,7-bisphosphate 7-phosphatase, EC 3.1.3.82, KEGG K03273. **Direct evidence.**

Downstream users of the pathway product (evidence the module is functional; not part of M00064)

- **PP_0342** **waaC** (heptosyltransferase I, EC 2.4.99.23), **PP_0341** **waaF** (heptosyltransferase II, EC 2.4.99.24), **PP_0344** **waaP** (heptose-I kinase), **PP_0345** **WapP**, **PP_0346** **WapQ** — an intact inner-core cluster. Their presence is strong indirect evidence that ADP-L-glycero-D-manno-heptose is synthesized in KT2440.

Step 5 (HldD/GmhD epimerase) — no ortholog present (**gap**)

No KT2440 protein is a credible HldD. Evidence: aligning the entire KT2440 IPR001509 epimerase set (9 proteins) against the same-genus reference HldD (*P. aeruginosa* PA3337/Q9HYQ8, 330 aa, EC 5.1.3.20, carries HldD-specific IPR011912) yields a best identity of only **34.8% (PP_3129 gale)** — below even the *galE*↔*gale* cross-species value (44.1%) and far below true-ortholog calibration (*gmhA* 91.8%, *gmhB* 80.0%). No KT2440 protein is in UniRef50/90_Q9HYQ8. - **PP_3206 (Q88HZ6) — RETRACTED as HldD candidate.** KEGG K26183 defines it as *fdeJ* (naringenin/flavonoid degradation, map00946). Its PANTHER PTHR43103:SF3 assignment is shared with HldD only at the broad superfamily level; sequence identity to genuine HldD is ~33% (background). **Not the epimerase.** - Other epimerase-family proteins are their own enzymes, not HldD: **PP_3129 gale** and **PP_0501** are *GalE*-type (InterPro IPR005886 UDP_G4E); **PP_5305** and **PP_1803 wbpV** are in PANTHER PTHR48079; **PP_1805 wbpM** is a large polytopic protein. None carries the HldD-specific IPR011912.

Candidate-list genes that are OUT of scope (exclude from this module)

KDO: PP_0956 `kdsC` (K03270), PP_0957/PP_1806 (K06041 API), PP_1611/PP_1807 `kdsA1/2` (K01627), PP_1902 `kdsB` (K00979). dTDP-rhamnose: PP_0265 `rmlC`, PP_0500, PP_1782 `rfbC`, PP_1783 `rfbA`, PP_1784 `rfbD`, PP_1785 `rffG`. GDP-mannose/fucose: PP_1277/PP_1776 `algA`, PP_1799 `gmd` (K01711), PP_1800 `rmd` (K22252). UDP-GlcNAc: PP_5411 `glmU`, PP_1811 `rffE`. UDP-glucuronate: PP_3821 `galU`, PP_2926 `udg`. Misc oxidoreductases PP_5212 (K00523), PP_1805 `wbpM` (K24300).

5. Gaps, ambiguities, and likely over-annotations

Genuine gap (single): HldD/GmhD (K03274, EC 5.1.3.20). Confirmed absent by four independent methods: KEGG (no `ppu` K03274), UniProt/UniRef (no KT2440 member in UniRef50/90 of PA3337 HldD), NCBI RefSeq (no heptose-6-epimerase record for txid160488), and genome tblastn (best hit 29%, a `galE`, across all 10 KT2440 assemblies). KEGG lists M00064 COMPLETE only for *E. coli*, not *ppu*.

This is a real absence, not an annotation artifact — the earlier "annotation gap" hypothesis was disproven this iteration by the genome-level tblastn (an unannotated ORF would have been detected by 6-frame translation).

The biological puzzle and two remaining hypotheses: steps 1–4 (GmhA/HldE/GmhB) plus intact WaaC/WaaF/WaaG/WaaP mean ADP-heptose is made and transferred to the core, yet the D→L epimerase is missing. - **(a) D,D-heptose core (favored, testable):** KT2440 incorporates D-glycero-D-manno-heptose without epimerization; step 5 is then `not expected in target taxon` and the generic module boundary (which assumes the L,D-Hep endpoint) is wrong for this organism. Requires the L-vs-D stereochemistry of KT2440 core heptose (LPS structural data). - **(b) Non-orthologous epimerase:** a novel/analogous enzyme performs the D→L step; would warrant a new gene model and GO request.

Species-transfer caveat: *P. aeruginosa* (congener) uses L-glycero-D-manno-heptose in its inner core and *does* encode HldD (PA3337) — so the KT2440 absence is lineage/strain-specific and cannot be assumed from *Pseudomonas*-level evidence. The Helander et al. 1980 (

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heptose data are for "*P. putida*" generically and do not resolve heptose stereochemistry or strain.

Retracted (Iteration 2): PP_3206 is *fdeJ* (K26183, flavonoid degradation), not HldD; the Iteration-1 PANTHER-subfamily match was superfamily-level only. No KT2440 NAD-epimerase (PP_0501, PP_3206, PP_5305, PP_1803 wbpV, PP_3129 galE) is the epimerase.

Over-annotation / over-propagation risks for curation: - The 26-gene bucket massively over-includes; treating it as the module gene set would wrongly inflate coverage. - **PP_1323** **gmhA** carries a *diaA* synonym — do not let the DnaA-regulation annotation dilute its metabolic (GmhA) role. - Multiple generic "epimerase/dehydratase" proteins (PP_0501, PP_3206, PP_5305, PP_1803) risk being over-mapped to EC 5.1.3.20 without sequence proof. **PP_3206 in particular must NOT be mapped to HldD** — it is *fdeJ* (flavonoid degradation, K26183). None of these is the epimerase; the real HldD is absent from the reference proteome.

6. Module and GO-curation recommendations

Module step	Recommended status
GmhA (step 1)	covered (PP_1323)
HldE kinase (step 2)	covered (PP_4934)
GmhB (step 3)	covered (PP_0059)
HldE adenylyltransferase (step 4)	covered (PP_4934)
HldD/GmhD epimerase (step 5)	gap → module_needs_revision ; candidate not_expected_in_target_taxon pending LPS data

Actions: 1. **Redefine the module gene set** to the M00064 core (gmhA, hldE, gmhB, hldD) and explicitly exclude KDO/rhamnose/mannose/GlcNAc/glucuronate genes; the current **ppu00541** bucket boundary is wrong for this organism-specific module. 2. **Do NOT assign any gene to HldD in KT2440.** PP_3206 is *fdeJ* (K26183, flavonoid degradation). No KT2440 protein should be mapped to K03274/EC 5.1.3.20/GO:0008712 — the gene is genome-confirmed absent. 3. **Mark M00064 as** **module_needs_revision** **for PSEPK** and add an organism note that the terminal 6-epimerase is absent while steps 1–4 and the downstream heptosyltransferases are present. 4. **Request expert/experimental clarification of KT2440 inner-core heptose stereochemistry** (L- vs D-glycero-D-manno-heptose). If D,D-Hep, set step 5 **not_expected_in_target_taxon** and

create a KT2440-specific module variant ending at ADP-D,D-heptose. If L,D-Hep, open a search for a **non-orthologous epimerase** and a possible new gene/GO request. 5. Keep GO terms for steps 1–4 as-is (well-supported by UniProt EC annotations).

7. Genes to promote to full fetch-gene review

1. **PP_0342** waaC / **PP_0341** waaF (**highest priority, now**): the heptosyltransferases define which heptose stereoisomer is used. A fetch-gene review of their substrate specificity (vs characterized L,D-Hep transferases) is the fastest bioinformatic route to decide whether an L→D epimerase (HldD) is even required — directly resolving the step-5 gap without wet-lab work.
2. **PP_3206** fdeJ — review only to formally *exclude* it from HldD (confirm K26183 flavonoid-degradation assignment); prevents future over-mapping to EC 5.1.3.20.
3. **PP_4934** hldE — verify both catalytic domains are intact (single bifunctional protein carrying two module steps).
4. **PP_1323** gmhA — confirm the metabolic (isomerase) assignment is not overshadowed by the *diaA* regulatory annotation.
5. Note: there is **no HldD locus to promote** — genome tblastn confirms none exists.

8. Key references

- Kneidinger B, et al. *Biosynthesis pathway of ADP-L-glycero-β-D-manno-heptose in Escherichia coli*. J Biol Chem 2002.

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— Defines the GmhA/HldE/GmhB/HldD four-enzyme pathway.

- McArthur F, et al. *Functional analysis of the glycero-manno-heptose 7-phosphate kinase domain from the bifunctional HldE protein...* J Bacteriol 2005.

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— HldE is bifunctional (kinase + adenylyltransferase).

- Desroy N, et al. *Towards Gram-negative antivirulence drugs: new inhibitors of HldE kinase*. 2009.

P 19124251

— HldE bifunctional role; heptose-less LPS attenuates virulence.

- Helander I, et al. *Chemical structure and inhalation toxicity of lipopolysaccharides...* 1980.

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— *P. putida* LPS contains heptose.

- KEGG module **M00064**, orthology links rest.kegg.jp/link/ppu/K03271...K03274, and KO **K26183** (fdeJ) definition (accessed 2026-08-08).
- UniProt proteome **UP000000556** (organism 160488); reference HldD **Q9HYQ8** (*P. aeruginosa* PA3337, EC 5.1.3.20, InterPro IPR011912) and *E. coli* HldD **P67910**; **UniRef50/UniRef90_Q9HYQ8** cluster membership (no taxon-160488 member) (accessed 2026-08-08).
- **NCBI**: RefSeq protein / Gene eSearch for txid160488 (no *hldD/gmhD/rfaD* or heptose-6-epimerase); **tblastn** of Q9HYQ8 vs KT2440 genome **AE015451.2** (+ assemblies CP167163.1, CP097524-529, CP169744.1, CP159467.1) — best hit 29% id, E=9e-15 (accessed 2026-08-08).

Uncertainty & species-transfer statement

Steps 1–4 rest on **direct KT2440 evidence** (KEGG ortholog assignments + UniProt curated EC numbers for PP_1323/PP_4934/PP_0059) and are high-confidence. Step 5 (HldD) is now settled at the sequence/genome level as a **direct-for-KT2440 absence**: no gene in KEGG/UniProt/RefSeq, no UniRef50/90 membership, and genome tblastn (6-frame) finds only a 29% *galE*-level hit across all 10 KT2440 assemblies — calibrated against 80–92%-identity true PA↔PP orthologs. What remains genuinely open is the **biological interpretation**: (a) a D,D-heptose core with no epimerase requirement (→ [not_expected_in_target_taxon](#)), or (b) a non-orthologous epimerase. This is **inferred, not demonstrated**, and needs KT2440 LPS structural data (heptose stereochemistry) and/or heptosyltransferase substrate characterization. Species transfer is explicitly weak here: the congener *P. aeruginosa* uses L,D-Hep and *does* encode HldD (PA3337), so *Pseudomonas*-level or generic-bacterial evidence must **not** be used to assert an HldD in KT2440. The Helander et al. 1980

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heptose datum is species-generic and does not resolve stereochemistry or strain.

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